

InstructPix2Pix Pipeline-Stage Attribution

Controlled ablations of VAE reconstruction, latent delta-z, geometric perturbation, composition order, seeds, optimization depth, and loss terms

Valid primary cases	Editor seeds replayed	Iteration depths	Loss settings
3	6	1, 5, 10, 20, 30	3

Central attribution

Across the corrected runs, the largest learned effect comes from latent delta-z. VAE encode/decode reconstruction creates a smaller baseline shift. Learned geometry on the original is usually minor, while geometry after reconstruction can be amplified by diffusion seed sensitivity. The extreme black-jacket collapse appears only after deeper latent optimization and is not a geometry-only result.

All findings below use saved images and exact within-case prompt, seed, diffusion-step, guidance, SSIM, PSNR, L2, and ArcFace iResNet-100 records. Visual review takes precedence when a face-identity embedding conflicts with the visible result.

1. Controlled pipeline

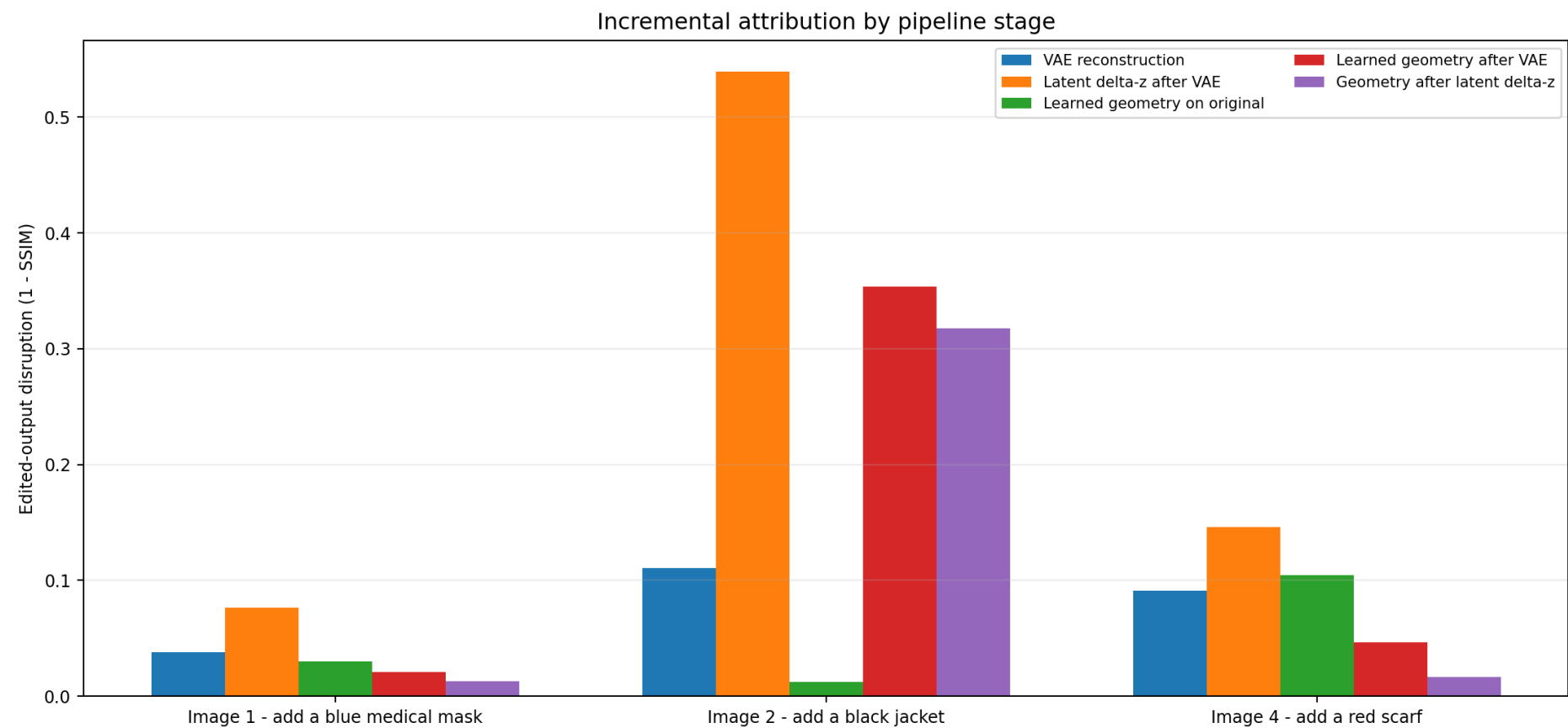
The stock branch edits the normalized original image. The ablation branches separately replay: VAE encode/decode reconstruction; optimized latent delta-z; learned geometry on the original; learned geometry on the reconstruction; the current combined order geometry(decode(encode(x) + delta-z)); and the order control decode(encode(geometry(x))). The editor settings are held fixed within a comparison.

Stage	Editor input	What it isolates
Stock	x	Reference edit
VAE reconstruction	decode(encode(x))	Encode/decode reconstruction effect
Latent delta-z	decode(encode(x) + delta-z)	Optimized latent contribution
Geometry on original	geometry(x)	Learned geometry without VAE reconstruction
Geometry after VAE	geometry(decode(encode(x)))	Geometry acting on reconstructed pixels
Combined	geometry(decode(encode(x) + delta-z))	Current latent plus geometry path
Order control	decode(encode(geometry(x)))	Sensitivity to operation order

Validity controls

The normalized 512x512 original saved inside each case is used for both stock editing and stage comparison. Neutral warps were audited at 512x512. Polar and TPS coordinate bugs were fixed before the corrected runs. A neutral grid-sample round trip can still change saved pixels by one 8-bit level and can therefore cause a small diffusion-output change; this is reported as a resampling floor, not as learned geometry.

2. Primary attribution summary



Case	VAE recon SSIM	Delta increment SSIM	Geometry-original SSIM	Geometry-after-VAE SSIM	Geometry-after-delta SSIM
Image 1 - add a blue medical mask	0.962	0.924	0.970	0.979	0.987
Image 2 - add a black jacket	0.889	0.461	0.988	0.646	0.683
Image 4 - add a red scarf	0.909	0.854	0.895	0.953	0.984

3.1 Image 1 - add a blue medical mask

Warp: tps | iterations: 20 | seed: 1234 | image guidance: 1.5

The clean baseline is already an imperfect mask edit. Delta-z changes texture and face coverage, while learned TPS adds little beyond the delta-z replay.

Latent delta-z replay

delta_only | learned_delta_only | Add a blue medical mask

Original



Stage input



Clean edit



Stage edit



Independent geometry replay

geometry_only_original | learned_geometry_on_original | Add a blue medical mask

Original



Stage input



Clean edit



Stage edit



3.1.1 Incremental metrics

Increment	Mode	Edit SSIM	Edit PSNR	Edit L2	ArcFace edit similarity
VAE reconstruction	baseline_control	0.962	27.01	0.045	0.946
Latent delta-z after VAE	delta_only	0.924	24.15	0.062	0.370
Learned geometry on original	geometry_only_original	0.970	29.22	0.035	0.974
Learned geometry after VAE	geometry_only_reconstruction	0.979	28.39	0.038	0.973
Geometry after latent delta-z	combined	0.987	34.70	0.018	0.995
Operation-order change	combined	0.922	24.50	0.060	0.347

Combined latent delta-z plus geometry

combined | learned_combined | Add a blue medical mask

Original



Stage input



Clean edit



Stage edit



3.2 Image 2 - add a black jacket

Warp: lens | iterations: 30 | seed: 24001 | image guidance: 2.0

The clearest transition: latent delta-z produces a seed-sensitive hood-like collapse after 20-30 iterations. Geometry on the original is negligible.

Latent delta-z replay

delta_only | learned_delta_only | Add a black jacket

Original



Stage input



Clean edit



Stage edit



Independent geometry replay

geometry_only_reconstruction | learned_geometry_on_reconstruction | Add a black jacket

Original



Stage input



Clean edit



Stage edit



3.2.1 Incremental metrics

Increment	Mode	Edit SSIM	Edit PSNR	Edit L2	ArcFace edit similarity
VAE reconstruction	baseline_control	0.889	18.65	0.117	0.833
Latent delta-z after VAE	delta_only	0.461	10.15	0.311	0.280
Learned geometry on original	geometry_only_original	0.988	42.34	0.008	0.994
Learned geometry after VAE	geometry_only_reconstruction	0.646	15.57	0.167	0.663
Geometry after latent delta-z	combined	0.683	12.68	0.232	0.399
Operation-order change	combined	0.501	12.35	0.241	0.328

Combined latent delta-z plus geometry

combined | learned_combined | Add a black jacket

Original



Stage input



Clean edit



Stage edit



3.3 Image 4 - add a red scarf

Warp: polar | iterations: 25 | seed: 1234 | image guidance: 1.5

The requested scarf remains visible. Delta-z mainly changes texture and color; corrected polar geometry modestly changes accessory placement.

Latent delta-z replay

delta_only | learned_delta_only | Add a red scarf

Original



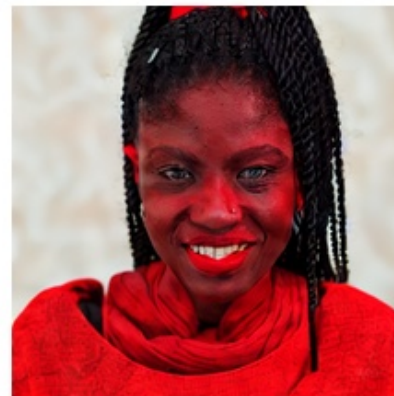
Stage input



Clean edit



Stage edit



Independent geometry replay

geometry_only_original | learned_geometry_on_original | Add a red scarf

Original



Stage input



Clean edit



Stage edit



3.3.1 Incremental metrics

Increment	Mode	Edit SSIM	Edit PSNR	Edit L2	ArcFace edit similarity
VAE reconstruction	baseline_control	0.909	28.20	0.039	0.851
Latent delta-z after VAE	delta_only	0.854	26.06	0.050	-0.027
Learned geometry on original	geometry_only_original	0.895	26.46	0.048	0.788
Learned geometry after VAE	geometry_only_reconstruction	0.953	31.09	0.028	0.915
Geometry after latent delta-z	combined	0.984	38.71	0.012	0.994
Operation-order change	combined	0.840	25.11	0.056	-0.014

Combined latent delta-z plus geometry

combined | learned_combined | Add a red scarf

Original



Stage input



Clean edit

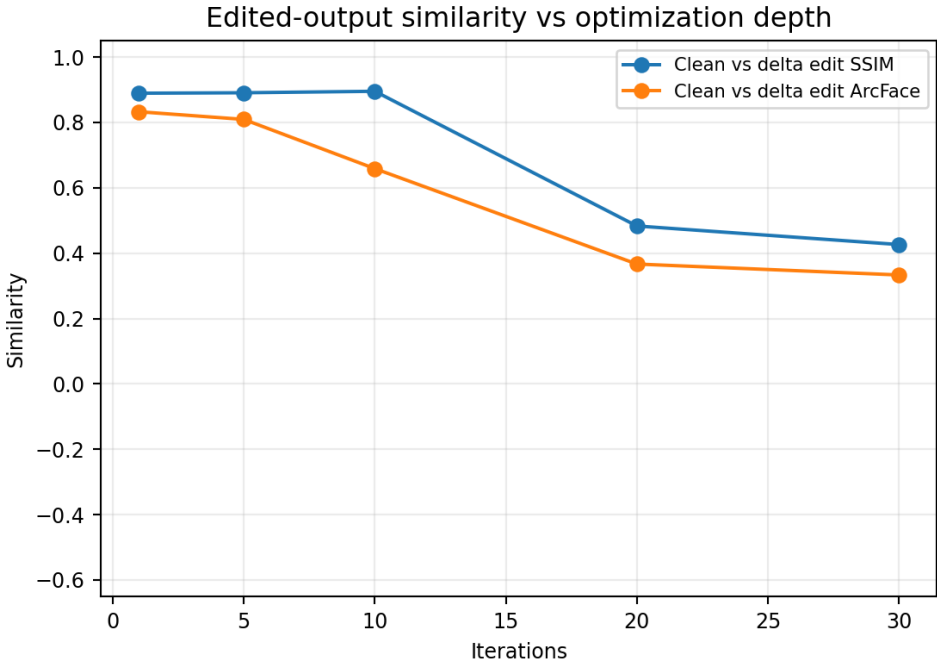
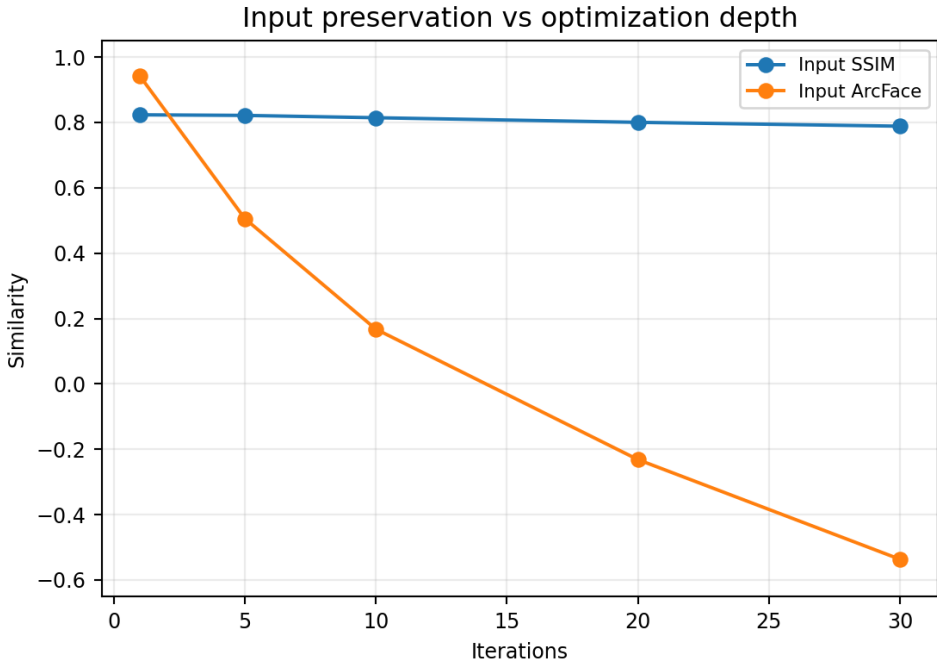


Stage edit



4. Optimization-depth control

Image 2, black-jacket prompt, seed 24001, delta-z only. The 20-iteration point is where the saved edit changes abruptly from a recognizable jacket edit to a hood-like collapse. Input identity similarity has already degraded strongly by 10 iterations; input SSIM continues to decline more gradually.



Iterations	Final loss	Input SSIM	Input ArcFace	Clean vs delta edit SSIM	Clean vs delta edit ArcFace
1	0.264	0.823	0.941	0.889	0.833
5	-4.086	0.821	0.506	0.891	0.809
10	-7.421	0.814	0.167	0.895	0.659
20	-11.353	0.800	-0.232	0.483	0.367
30	-14.345	0.788	-0.537	0.426	0.333

4.1 Visual transition across iteration depth

5 iterations

delta_only | learned_delta_only | Add a black jacket

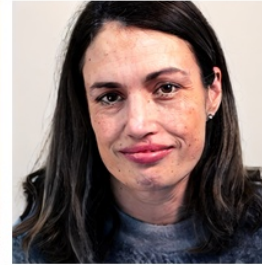
Original



Stage input



Clean edit



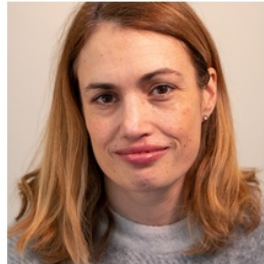
Stage edit



10 iterations

delta_only | learned_delta_only | Add a black jacket

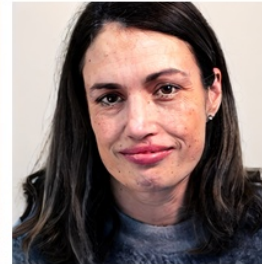
Original



Stage input



Clean edit



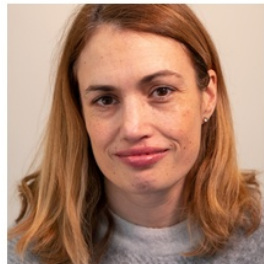
Stage edit



20 iterations

delta_only | learned_delta_only | Add a black jacket

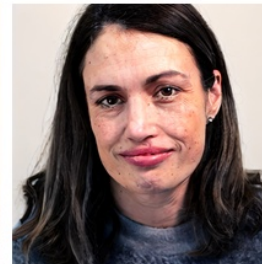
Original



Stage input



Clean edit



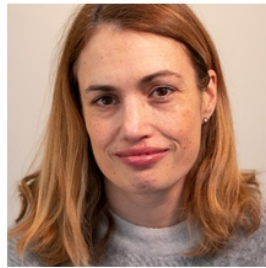
Stage edit



30 iterations

delta_only | learned_delta_only | Add a black jacket

Original



Stage input



Clean edit

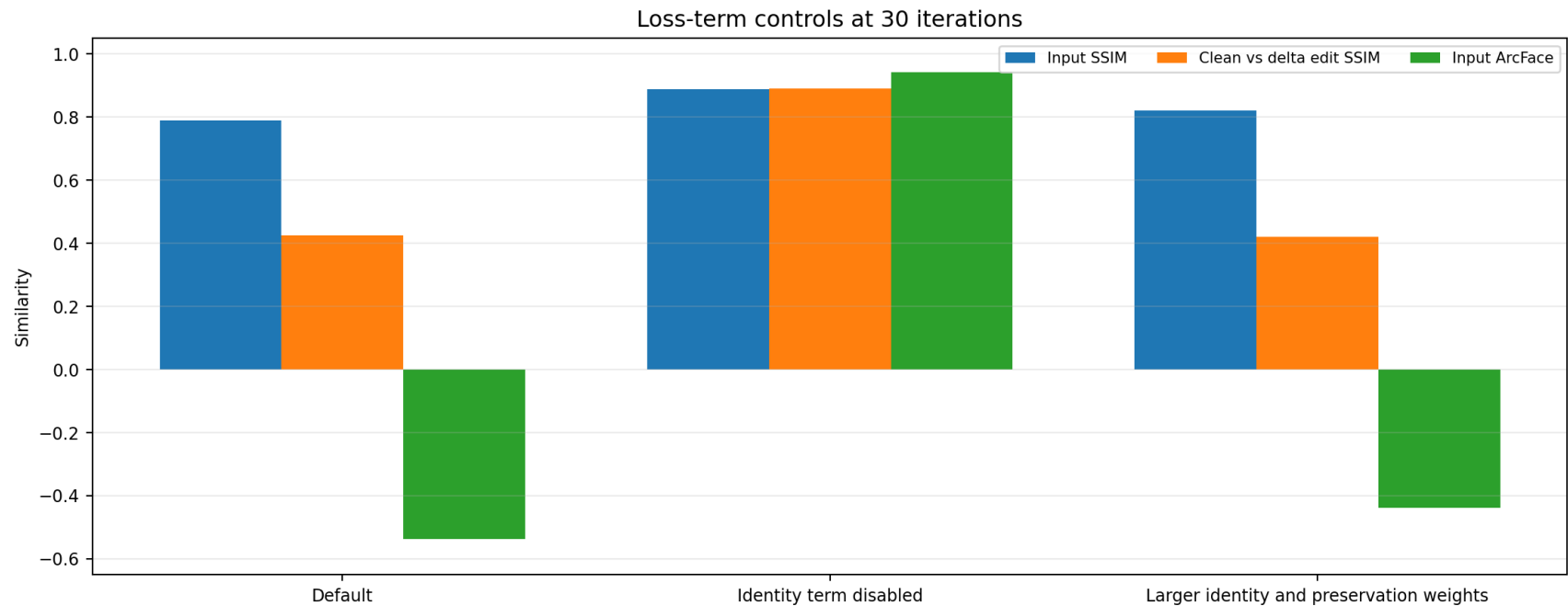


Stage edit



5. Loss-term controls

The optimization loss includes a negative identity-distance term, so a larger identity weight rewards identity disruption rather than preservation. The no-identity control removes that driver. The third control increases both the identity-disruption weight and the SSIM/pixel penalties; its high identity weight still dominates and produces collapse.



Setting	Identity w.	SSIM w.	Pixel w.	Input SSIM	Input ArcFace	Edit SSIM	Edit ArcFace
Default	10.0	5.0	0.1	0.788	-0.537	0.426	0.333
Identity term disabled	0.0	5.0	0.1	0.888	0.943	0.891	0.954
Larger identity and preservation weights	10.0	20.0	1.0	0.821	-0.438	0.420	0.304

Result

With identity weight set to zero, the 30-iteration output remains close to the VAE-reconstruction edit (clean-vs-stage edit SSIM 0.891; ArcFace 0.955). This isolates the identity-disruption objective as the cause of the large latent change in the default run.

5.1 Loss-control image comparison

Identity term disabled

delta_only | learned_delta_only | Add a black jacket

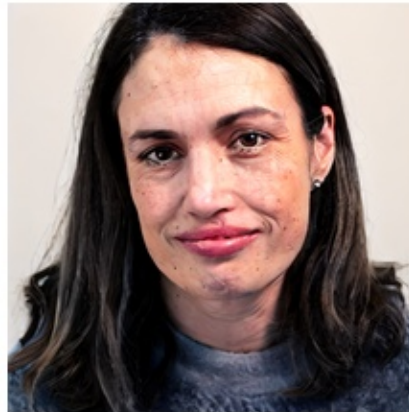
Original



Stage input



Clean edit



Stage edit



Larger identity and preservation weights

delta_only | learned_delta_only | Add a black jacket

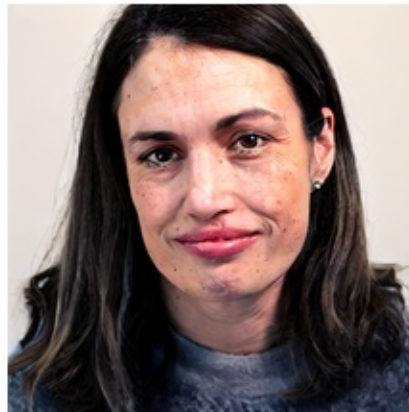
Original



Stage input



Clean edit



Stage edit



6. Seed sensitivity

Saved stage inputs were replayed with three diffusion seeds without rerunning optimization. Image 2 shows a consistent delta-z effect, but the extreme hood collapse is concentrated at seed 24001. Image 4 keeps the requested scarf across all seeds; its low or negative ArcFace values conflict with the visible face and are treated as an embedding failure under global recoloring, not as visual proof of identity loss.

Image 2 - black jacket

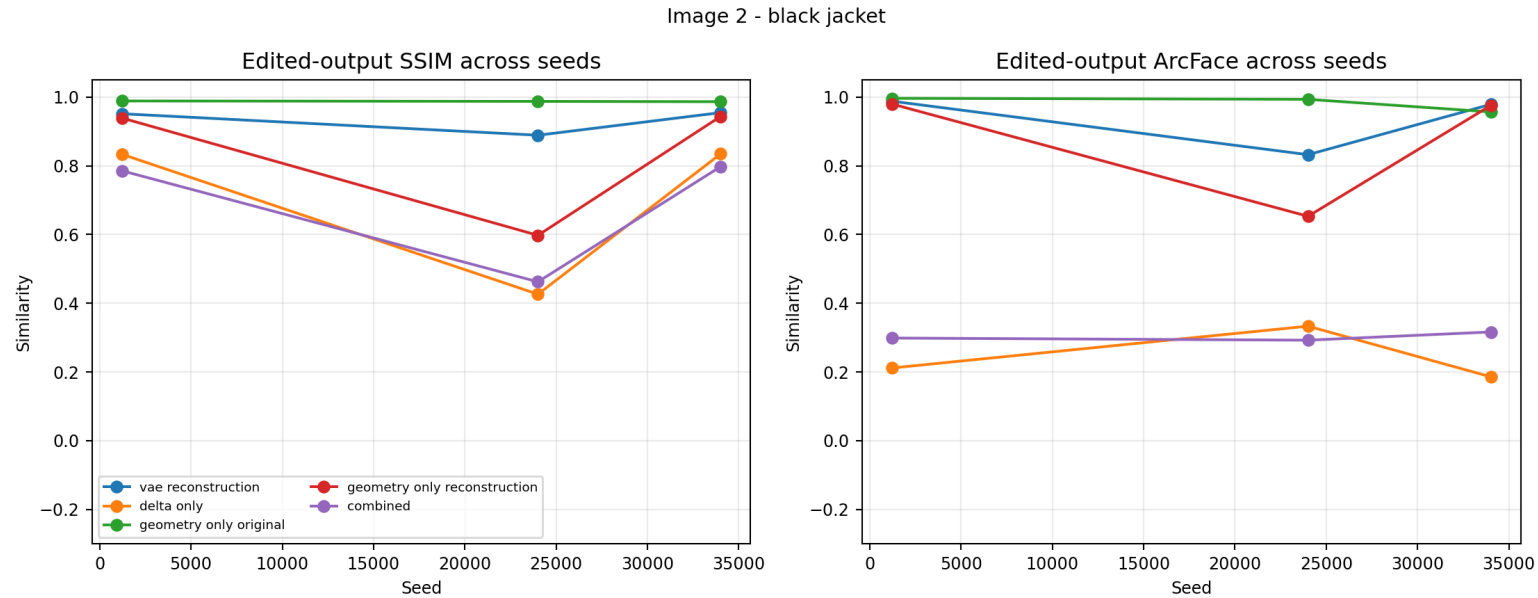
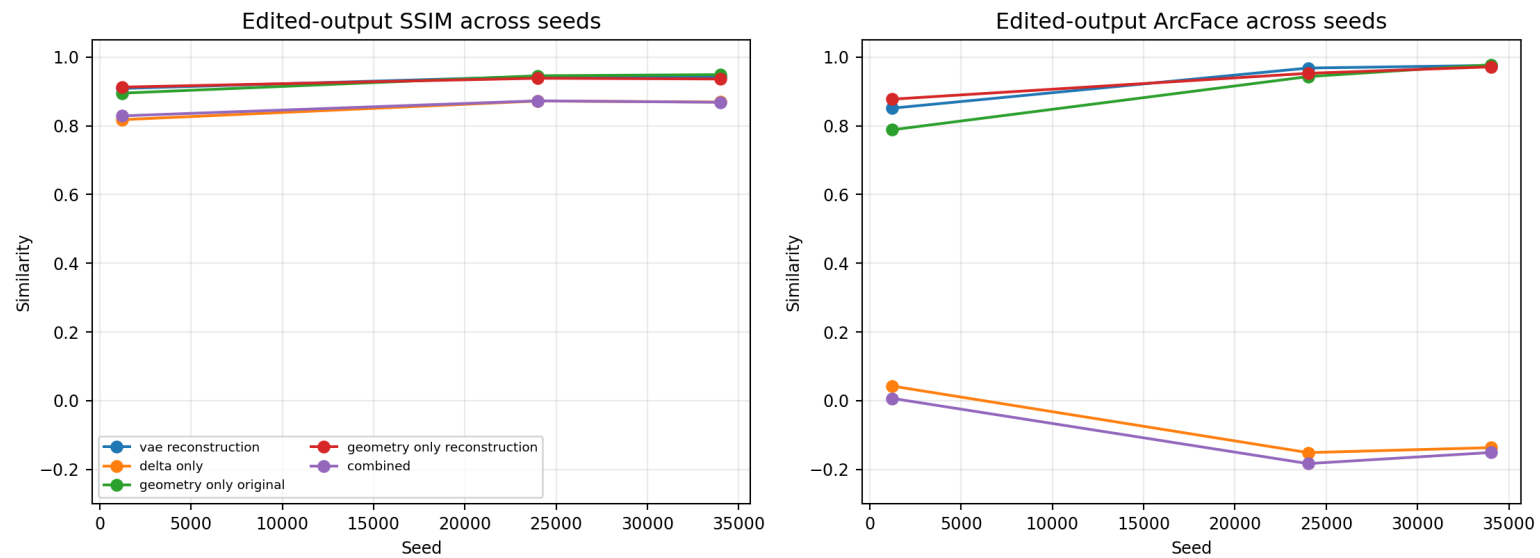


Image 4 - red scarf

Image 4 - red scarf



6.1 Representative seed replays - Image 2 - black jacket

Seed 1234 - latent delta-z

seed=1234 | delta_only | Add a black jacket

Original



Stage input



Clean edit



Stage edit



Seed 24001 - latent delta-z

seed=24001 | delta_only | Add a black jacket

Original



Stage input



Clean edit



Stage edit



Seed 34007 - latent delta-z

seed=34007 | delta_only | Add a black jacket

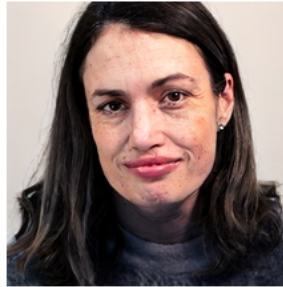
Original



Stage input



Clean edit



Stage edit



6.1 Representative seed replays - Image 4 - red scarf

Seed 1234 - latent delta-z

seed=1234 | delta_only | Add a red scarf

Original



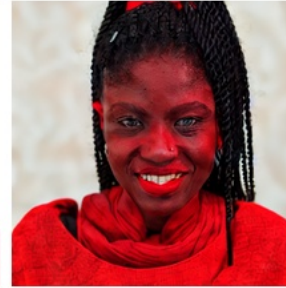
Stage input



Clean edit



Stage edit



Seed 24001 - latent delta-z

seed=24001 | delta_only | Add a red scarf

Original



Stage input



Clean edit



Stage edit



Seed 34007 - latent delta-z

seed=34007 | delta_only | Add a red scarf

Original



Stage input



Clean edit



Stage edit



7. Correctness caveats and excluded runs

The report does not average every DONE folder. It uses an explicit valid-run manifest and records why superseded results are excluded.

Excluded folder	Reason
pilot3_image1_blue_mask_bspline_10	Excluded: clean edit used a 256x256 source while ablation stages used a 512x512 normalized source.
image1_blue_mask_tps_20_corrected	Excluded: superseded by corrected_v2 after the same clean-baseline resolution mismatch was fixed.
image4_red_scarf_polar_25	Excluded: zero-state polar warp was not an identity transform; replaced by the corrected polar run.

Neutral-warp audit

After correction, all 13 audited warps passed at 512x512. TPS and polar zero states now produce identity-coordinate grids to floating-point tolerance. Saving a neutral grid-sampled image to 8-bit PNG can still create one-level pixel changes, so neutral edit differences are retained as resampling controls.

Metric caveat

ArcFace iResNet-100 is informative for ordinary face crops but can become unreliable when the edit recolors or occludes most of the face. Negative ArcFace similarity is therefore not called an identity failure unless it agrees with the visible images. SSIM and PSNR are also similarity diagnostics, not semantic success labels.

8. Evidence-backed conclusions

1. VAE reconstruction alone causes a measurable but usually moderate edit shift.
- 2. Latent delta-z is the dominant optimized source of the observed InstructPix2Pix changes in the corrected cases.**
- 3. Learned geometry on the original is usually close to the neutral-resampling control; it does not explain the black-jacket collapse.**
4. Geometry after reconstruction can be amplified by editor sensitivity, but its effect is more seed-dependent and smaller than delta-z in the strongest case.
- 5. The large black-jacket collapse begins between 10 and 20 optimization iterations and is most extreme at seed 24001.**
- 6. Removing the identity-disruption term prevents the large latent collapse at 30 iterations.**
7. The strongest output disruptions coincide with visible input degradation, so they are not evidence of an imperceptible geometric attack.
8. The red-scarf and blue-mask prompts remain broadly successful; their metric changes are not convincing edit failures.

Final attribution

The observed InstructPix2Pix edit changes are primarily produced by optimized latent delta-z, with smaller baseline effects from VAE reconstruction and context-dependent amplification from reconstruction plus resampling or geometry. The current evidence does not support learned geometry alone as the main cause.

Artifacts collected from: C:\Users\parth\Desktop\experiment\amit_code\targeted_experiments\stage_attribution